Exact Madelung and Bohm Quantum Potential Identities in Core Lean 4 (Zero Mathlib Dependencies)

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Abstract

We provide a standalone, zero-`sorry` formalization in core Lean 4 (`BohmIdentities.lean`, requiring zero external Mathlib imports and verifying directly with `lean BohmIdentities.lean`) of the cleared algebraic identities underlying the Madelung hydrodynamical transform and Bohmian guidance law. Given polar components $\psi = (Rc, Rs)$ and product-rule differential $d\psi = (dR\,c - R\,s\,dS,\ dR\,s + R\,c\,dS)$, the theorem `Bohm.guidance_cleared` proves $\mathrm{Im}(d\psi \cdot \overline{\psi}) = dS \cdot |\psi|^2$, and `Bohm.quantumPotential_cleared` proves the cleared prefactor identity $(-\Delta R)(2R) = 2((-\Delta R)R)$ for the Bohm quantum potential $Q = -\frac{\hbar^2}{2m}\frac{\Delta R}{R}$.

1 Formalized Theorems in `BohmIdentities.lean`

Theorem 1 (`Bohm.guidance_cleared`)**.** *For all $R, dR, c, s, dS \in \mathbb{Z}$:*

$$(dR \cdot s + R \cdot c \cdot dS)(R \cdot c) - (dR \cdot c - R \cdot s \cdot dS)(R \cdot s) = dS \cdot \big((R \cdot c)^2 + (R \cdot s)^2\big).$$

Theorem 2 (`Bohm.quantumPotential_cleared`)**.** *For all $\Delta R, R \in \mathbb{Z}$:*

$$(-\Delta R) \cdot (2R) = 2 \cdot ((-\Delta R) \cdot R).$$

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